Calibration of Probabilities

Michele Davide Buonsante

Università degli Studi di Milano

I declare that this material, which I now submit for assessment, is entirely my own work and has not been taken from the work of others, save and to the extent that such work has been cited and acknowledged within the text of my work. I understand that plagiarism, collusion, and copying are grave and serious offences in the university and accept the penalties that would be imposed should I engage in plagiarism, collusion or copying. This assignment, or any part of it, has not been previously submitted by me or any other person for assessment on this or any other course of study.

Abstract

This project studies the calibration of probabilistic classifiers. We evaluate Logistic Regression and Random Forest, apply Platt scaling and Isotonic regression to improve probability estimates. The models are tested on two real-world datasets: Adult and Bank.

1 Introduction

In supervised learning, many models of probabilistic classifier output probabilities rather than hard decisions, giving us a value between 0 and 1 representing the estimated likelihood of belonging to a certain class.

However, since these probabilities are not always reliable, we introduce calibration that refers to the ability of a model to produce reliable probability estimates. A model may achieve high classification accuracy while still providing poorly calibrated probability estimates. It measures how much these probabilities correspond to real statistical frequency of the events in the real world. High accuracy does not necessarily imply good calibration, since calibration concerns the quality of predicted probabilities rather than classification correctness [1].

In this project we study the calibration of probabilistic classifiers, considering two models, Logistic Regression and Random Forest, both implemented from scratch. Then, we apply post-processing calibration methods, Platt scaling and Isotonic regression.

The performance will be evaluated using Accuracy, Log-loss, Brier score and with representation of Reliability diagrams.

2 Datasets

We used two real-world binary classification datasets, one obtained from OpenML and the other from the `sklearn.datasets` library. The two datasets were intentionally selected with different characteristics in order to analyze how calibration methods behave under different data conditions.

The first dataset is the Adult dataset, based on the American census of 1994, containing a mix of numerical and categorical data, with the objective of predicting if a person earns more than 50k dollars per year. This dataset contains a relatively large number of samples and is more balanced compared to the second dataset.

The second dataset used in this project is the Breast Cancer Wisconsin dataset. It is a binary classification dataset containing numerical features computed from digitized images of breast mass cell nuclei. The objective is to classify tumors as malignant or benign. Compared to the Adult dataset, this dataset is significantly smaller, containing 569 samples. This characteristic makes it particularly useful for analyzing the behavior of calibration methods in settings with limited calibration data.

3 Methodology

3.1 Data preprocessing

Both datasets were preprocessed before training. Numerical features were standardized using a standard scaler, while categorical features were encoded using one-hot encoding.

Each dataset was split into training, validation, and test sets. The training set was used to fit the models, the validation set was used for calibration, and the test set was used for final evaluation. This separation has been important in order to avoid data leakage. A model is perfectly calibrated if:

$$P(Y = 1 \mid \hat{P} = p) = p$$

3.2 Models

We implemented two classification models from scratch:

- **Logistic Regression**: a linear model that outputs probabilities through a sigmoid function.

  $$\sigma(z) = \frac{1}{1 + e^{-z}}$$

  The model was trained using gradient descent.

- **Random Forest**: an ensemble method composed of multiple decision trees trained on bootstrap samples of the data. Each tree is built using a random subset of features, and predictions are obtained by majority voting. Probabilities are estimated as the average of predictions across trees.

3.3 Calibration methods

To improve the quality of probability estimates, we applied two post-processing calibration techniques:

- **Platt scaling**: a method that fits a logistic regression model on the predicted probabilities of the base model, transforming them using a sigmoid function.

$$P(y = 1 \mid x) = \frac{1}{1 + \exp(Af(x) + B)}$$

- **Isotonic regression**: a non-parametric method that fits a monotonic function to map predicted probabilities to calibrated values. In this project, isotonic regression was implemented using the Pool Adjacent Violators (PAV) algorithm. It starts with one block for each validation example and repeatedly merges adjacent blocks whenever their average labels violate the monotonicity constraint.

Both calibration models were trained using the validation set.

3.4 Evaluation metrics

We evaluated model performance using the following metrics:

- **Accuracy**: measures the proportion of correctly classified samples.
- **Log Loss**: evaluates the quality of predicted probabilities, penalizing confident but incorrect predictions.

$$\text{LogLoss} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log(p_i) + (1 - y_i)\log(1 - p_i)\right]$$

- **Brier Score**: measures the mean squared error between predicted probabilities and true labels.

$$\text{Brier} = \frac{1}{N}\sum_{i=1}^{N}(p_i - y_i)^2$$

In addition, we used **reliability diagrams** to visually assess calibration. These plots compare predicted probabilities with the observed frequency of the positive class. A perfectly calibrated model corresponds to a diagonal line in the reliability diagram.

4 Results

4.1 Adult Dataset

Model	Accuracy	Log Loss	Brier
LR	0.8490	0.3245	0.1035
LR + Platt	0.8447	0.3663	0.1112
LR + Iso	0.8496	0.3349	0.1038
RF	0.8274	2.9070	0.1396
RF + Platt	0.8274	0.4284	0.1313
RF + Iso	0.8274	0.4108	0.1263

Table 1: Results on the Adult dataset.

4.2 Breast Cancer Dataset

Model	Accuracy	Log Loss	Brier
LR	0.9737	0.0767	0.0211
LR + Platt	0.9825	0.1196	0.0243
LR + Iso	0.9474	1.8179	0.0526
RF	0.9386	0.1361	0.0446
RF + Platt	0.9386	0.1742	0.0441
RF + Iso	0.9474	0.3885	0.0378

Table 2: Results on the Breast Cancer dataset.

4.3 Reliability Diagrams

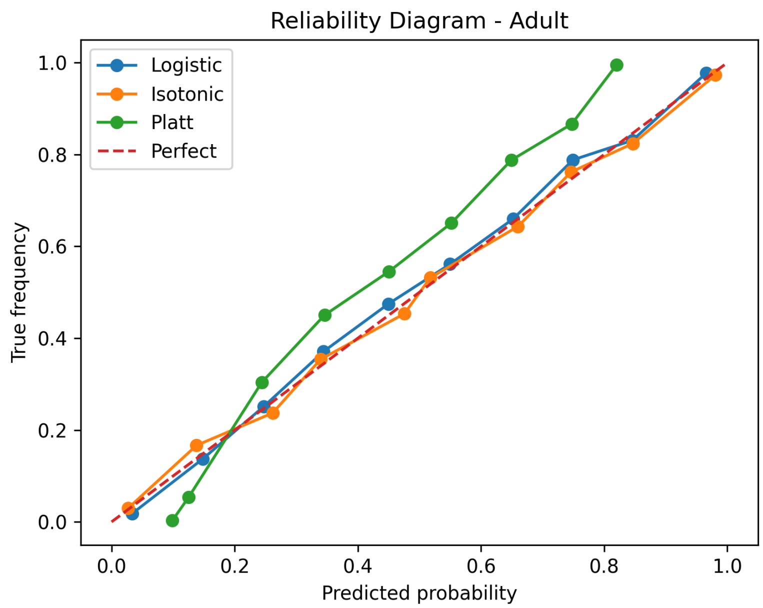

Figure 1: Calibration curves for Logistic Regression on the Adult dataset.

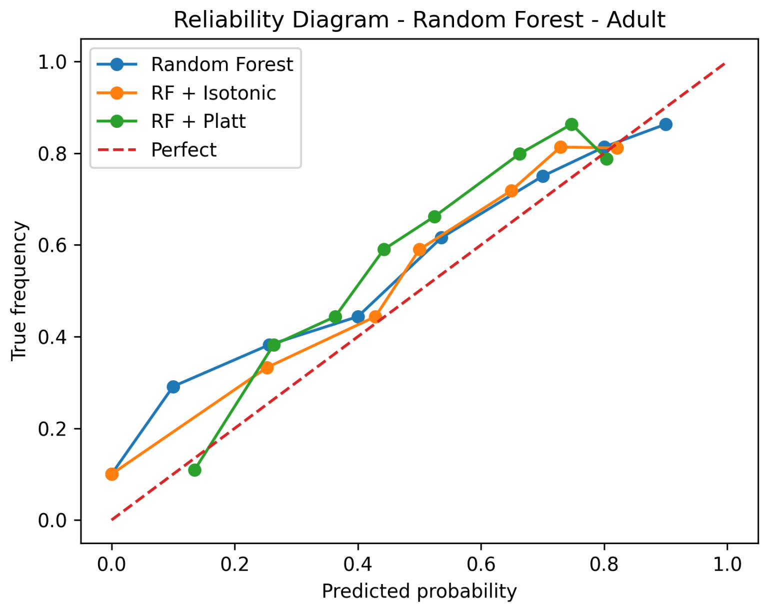

Figure 2: Calibration curves for Random Forest on the Adult dataset.

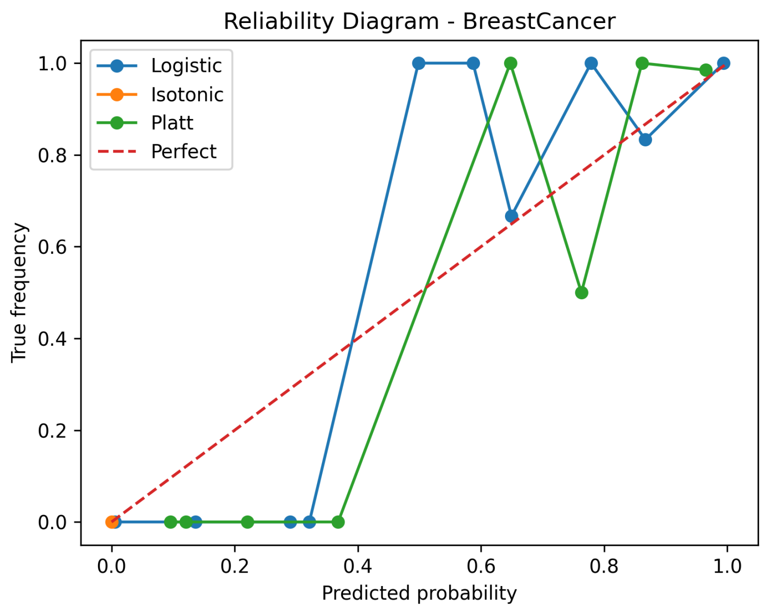

Figure 3: Calibration curves for Logistic Regression on the Breast Cancer dataset.

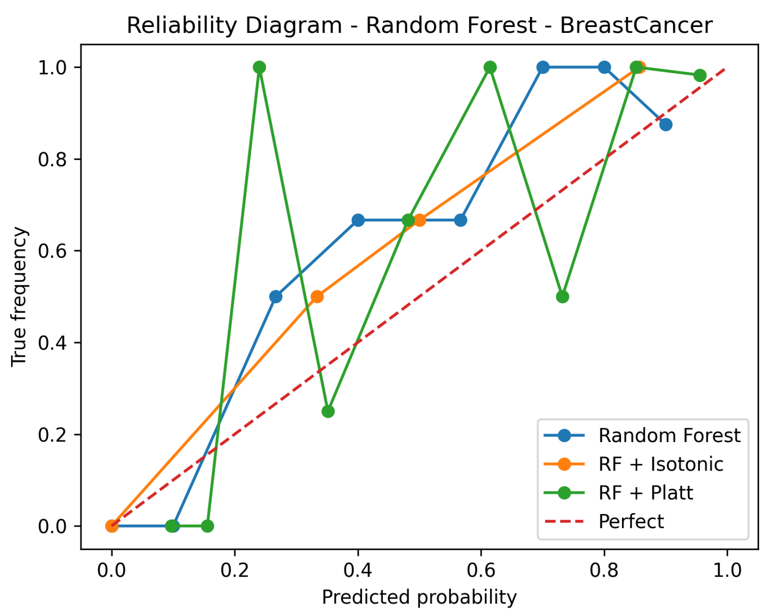

Figure 4: Calibration curves for Random Forest on the Breast Cancer dataset.

5 Analysis

From the results, we can observe that Logistic Regression tends to produce well-calibrated probabilities because it directly optimizes the log-likelihood, which corresponds to minimizing a proper scoring rule.

On the Adult dataset, the reliability diagrams show predictions relatively close to the diagonal line, indicating good calibration. In addition, the quantitative metrics remain stable even after applying calibration methods.

Applying Platt scaling to Logistic Regression slightly worsens performance in terms of Log Loss and Brier Score. This suggests that the model was already well calibrated, and the additional transformation introduces unnecessary distortion. This behavior is expected, as Logistic Regression is known to produce well-calibrated probabilities by design.

Isotonic regression produces relatively small changes on the Adult dataset, slightly improving some metrics while worsening others. This indicates that a flexible non-parametric calibration method can slightly refine probability estimates when calibration is already reasonably good.

The behavior is different on the Breast Cancer dataset. Although Logistic Regression achieves very strong quantitative results, the reliability diagrams appear irregular and unstable. This is likely caused by the limited size of the dataset, which contains only 569 samples. Since reliability diagrams estimate calibration by grouping probabilities into bins, having few samples per bin can produce noisy calibration curves and sudden variations in the observed frequencies.

The instability caused by limited calibration data becomes even more evident with isotonic regression. On the Breast Cancer dataset, isotonic regression significantly worsens Log Loss and Brier Score, suggesting strong overfitting. This behavior is consistent with the literature, where isotonic regression is known to require larger calibration datasets in order to remain stable and reliable.

Although the reliability diagrams do not show extremely large deviations from the diagonal line, Random Forest still produces very high Log Loss values on the Adult dataset. This suggests the presence of a small number of highly confident but incorrect predictions, which are heavily penalized by Log Loss. This behavior can be explained by the averaging mechanism of ensemble methods, which tends to push probability estimates toward the center, as said in Alexandru Niculescu-Mizil and Rich Caruana's paper [2]. Specifically, since a Random Forest computes the final probability as the arithmetic mean of individual tree predictions, it is statistically unlikely for all trees to converge on extreme values (0 or 1) simultaneously.

Platt scaling significantly improves calibration for Random Forest, drastically reducing Log Loss and improving the Brier Score. This shows that a simple parametric transformation can effectively correct overconfident or biased probability estimates.

Isotonic regression also improves some Random Forest metrics, particularly the Brier Score, but the improvements are less consistent compared to Platt scaling. On the smaller Breast Cancer dataset, isotonic regression again appears more unstable, confirming its sensitivity to limited calibration data.

Overall, these results confirm that calibration performance strongly depends both on the base model and on the amount of available calibration data. Logistic Regression tends to produce reasonably calibrated probabilities by design, while Random Forest benefits more from post-processing calibration methods. Furthermore, isotonic regression appears to require larger calibration datasets in order to avoid overfitting and maintain stable probability estimates.

6 Conclusion

In this project, we studied the calibration of probabilistic classifiers using Logistic Regression and Random Forest on two real-world datasets with different characteristics.

The results show that Logistic Regression generally provides reasonably calibrated probabilities by design, while Random Forest tends to produce less reliable probability estimates despite achieving competitive classification accuracy.

Among the calibration methods considered, Platt scaling proved to be the most stable and effective approach, particularly for Random Forest, where it significantly improved Log Loss and Brier Score. Isotonic regression produced mixed results: while it occasionally improved some metrics, it also showed instability and signs of overfitting on the smaller Breast Cancer dataset.

The experiments also highlighted that calibration quality strongly depends on the amount of available calibration data. In particular, reliability diagrams become noisier on smaller datasets, making calibration analysis more difficult and unstable.

Overall, our findings confirm the importance of evaluating not only classification accuracy, but also the reliability of predicted probabilities, especially in applications where probability estimates are directly used for decision making.

7 Acknowledgements

The work was inspired by the paper "Predicting Good Probabilities with Supervised Learning" by Niculescu-Mizil and Caruana, as well as the blog post "Understanding Model Calibration - A Gentle Introduction", which provided useful insights into calibration techniques and their interpretation.

References

[1] M. Pavlovic, *Understanding Model Calibration: A Gentle Introduction and Visual Exploration of Calibration and the Expected Calibration Error (ECE)*, ICLR Blogposts, 2025.

[2] A. Niculescu-Mizil, R. Caruana, *Predicting Good Probabilities With Supervised Learning*, Proceedings of the 22nd International Conference on Machine Learning (ICML), 2005.